The Alaoglu–Erdős Conjecture at Weight Two

A Padé Height Audit, Rational-Transfer No-Go Theorems,
and a Directed-Rounding Extension through 2^{29}

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A nonduplicative continuation of the supplied unified report

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Abstract

Let $x \in \mathbb{R}$ and suppose that 2^x and 3^x are integers. The Alaoglu–Erdős conjecture asserts that x is an integer. The supplied 327-page unified report already reduces a hypothetical counterexample to an exact rank-two monoid, develops many determinant, valuation, dynamical, matrix, and operator formulations, and identifies the remaining wall as an unproved weight-two relation between logarithms. This report does not repeat that material.

The full conjecture is not proved here. Instead, the report gives a new and fairly sharp audit of the most relevant post-report development: Kawashima's May 2026 Padé criterion for products of polylogarithms. The central new theorem is a representation-independent upper bound for Kawashima's height defect. It shows that a successful weight-two specialization requires the polylogarithm arguments to have almost maximal *arithmetic efficiency*: their local smallness must be within a factor $1 - 1/(m + 1)^2$ of their average global height. This one inequality rigorously rules out the direct prime-logarithm points, high-root splitting, ordinary telescoping decompositions, and every unbounded construction inside a fixed S-unit group.

A second new theorem goes further. At rational points, Kawashima's criterion is incompatible with having even $\log 2$ in the rational span of the resulting logarithms. The proof combines the height-defect bound with a Cramer/Hadamard bound on a prime-valuation matrix. A tensor-rank corollary rules out the corresponding formal quadratic transfer of the entire Alaoglu–Erdős relation. Thus the current Padé theorem cannot settle the conjecture by any rational-point encoding of the natural kind; a successful continuation must exploit a genuinely new mixed-denominator or mixed-embedding cancellation mechanism.

Finally, the exact MPFR verifier reproduced in the supplied report was rerun, without source changes, on the two new bit ranges $[2^{27}, 2^{28})$ and $[2^{28}, 2^{29})$. All 402,653,182 non-powers of two were separated from the nearest integer by directed logarithmic inequalities. At the stated software trust level this extends the finite exclusion to $2^x < 2^{29}$, so any nonintegral counterexample must satisfy $x > 29$.

Contents

1 Executive verdict and claim ledger

OPEN / CONDITIONAL.
The requested unconditional proof was not obtained. As of the literature check performed for this report, the problem remains an open special case of the Four Exponentials Conjecture [1]. No step below is promoted from conditional or computational status to theorem status merely to produce a desired conclusion.

The work nevertheless changes the technical picture in three concrete ways.

1. It extracts from Kawashima's criterion an intrinsic, representation-free obstruction (Theorem 6.1) and turns it into an arithmetic-efficiency threshold.
2. It proves that the entire rational-point transfer strategy is impossible under that criterion (Theorems 8.2 and 8.3), not merely that several obvious choices of a common denominator fail.
3. It extends the independent finite scan by two full bits, from 2^{27} to 2^{29} (Section 13).

Statement	Status	Logical role
Projective Padé defect bound	Proved	Necessary condition for every specialization of Kawashima's weight-two criterion.
Arithmetic-efficiency threshold	Proved	Quantifies the exact high-efficiency region reached by the present theorem.
Direct, radical, and telescoping encodings fail	Proved	Eliminates the most natural representations of integer logarithms.
Rational-span transfer impossibility	Proved	Shows that no rational specialization satisfying the criterion can even span $\log 2$.
Formal quadratic-transfer impossibility	Proved	Extends the preceding no-go from factor-by-factor substitutions to the natural prime-valuation tensor of a counterexample.
Fixed-S exceptional finiteness	Proved from published effective S-unit approximation	Any fixed multiplicative group supplies at most effectively many candidate Padé tuples.
Near-unit unimodular form	Proved	Converts a counterexample into infinitely many exact determinant zeros near 1, but in the low-efficiency regime.
Scan through 2^{29}	Directed-rounding computation	Excludes nonintegral $x \leq 29$ at a disclosed software trust boundary.
Full Alaoglu–Erdős conjecture	Open	Still requires a genuinely new weight-two transcendence theorem.

2 Scope: what is imported and what is not repeated

Write

$$M = 2^x, \qquad A = 3^x.$$

The supplied unified report is taken as the source for the following previously established facts and constructions:

- rational x is automatically integral, while a nonintegral solution is transcendental;
- a counterexample gives the exact determinant relation

$$\log M \,\log 3 - \log A \,\log 2 = 0; \tag{1}$$

- conditional on one counterexample, all solutions form the free additive monoid $\{n + k\beta : n, k \geq 0\}$ for a unique least nonintegral generator β;
- the corresponding output pairs form the free multiplicative monoid generated by $(2, 3)$ and (M, A);
- the ordinary Four Exponentials Conjecture implies the desired conclusion, whereas the Six Exponentials Theorem does not close the two-by-two case;
- the two outputs of a nonintegral solution cannot both be 3-smooth;
- the existing directed-rounding audit covers $2^x < 2^{27}$.

The long determinant, valuation, Sturmian, matrix, operator, and formalization developments of the supplied report are not copied. Only the minimal statements needed to connect the new Padé analysis to the original problem are restated.

3 The exact weight-two object

3.1 Prime-valuation form

Let $\mathcal{P}$ be the set of rational primes and let

$$E = \mathbb{Q}^{(\mathcal{P})}$$

be the finite-support rational vector space with basis $(e_p)_{p\in\mathcal{P}}$. For a positive rational number r, write

$$\boldsymbol{\nu}(r) = (v_p(r))_{p\in\mathcal{P}} \in E, \qquad \lambda(y) = \sum_p y_p \log p.$$

Unique factorization gives an elementary but important fact:

$$\lambda : E \longrightarrow \mathbb{R} \quad \text{is injective as a } \mathbb{Q}\text{-linear map.} \tag{2}$$

Indeed, after clearing denominators, $\lambda(y) = 0$ says that a positive rational number is 1.

Put

$$u = \boldsymbol{\nu}(M), \qquad v = \boldsymbol{\nu}(A), \qquad a = e_2, \qquad b = e_3.$$

Then (1) is the vanishing at the point λ of the rational symmetric tensor

$$T = u \odot b - v \odot a \in \mathrm{Sym}^2(E), \qquad y \odot z := \frac{y \otimes z + z \otimes y}{2}. \tag{3}$$

This is the exact weight-two object to which a product-of-logarithms theorem must apply.

3.2 Rank of the counterexample tensor

The following elementary rank audit will be used later to prevent a hidden low-dimensional escape in a formal Padé transfer.

Lemma 3.1 (Full range on the support span)**.** *Assume that (M, A) comes from a nonintegral counterexample. Let*

$$S = \mathrm{span}_{\mathbb{Q}}\{u, v, a, b\}.$$

Then $\dim S \geq 3$, the tensor T of (3) has rank $\dim S$, and consequently its contraction range is exactly S.

Proof. If $\dim S = 2$, then u and v lie in $\operatorname{span}\{e_2, e_3\}$, so both M and A are 3-smooth. This is excluded by the three-base/Six Exponentials consequence proved in the supplied report.

If $\dim S = 4$, use the four independent linear forms u, b, v, a as coordinates on S^*. The quadratic form associated with T is

$$X_1X_2 - X_3X_4,$$

whose matrix is the direct sum of two nondegenerate hyperbolic planes. Its rank is 4.

Suppose $\dim S = 3$. The rank is at most 3, and $T \neq 0$: equality $u \odot b = v \odot a$ would force the two supporting planes to coincide, hence $\dim S \leq 2$. If the rank were 1, the rational quadratic form would have the shape $c\,\xi(r)^2$ with $c \in \mathbb{Q}^\times$ and $0 \neq r \in S$. Evaluating at λ and using (1) would give $\lambda(r) = 0$, contrary to (2).

It remains to exclude rank 2. The vectors u, v are independent: otherwise $\log M/\log A$ would be rational, whereas it equals $\log 2/\log 3$. The vectors a, b are also independent. Hence the annihilators of $\operatorname{span}\{u, v\}$ and $\operatorname{span}\{a, b\}$ are two distinct rational isotropic lines. The radical of a rank-two form on a three-dimensional space is only one-dimensional, so at least one of these isotropic lines is nonradical. The rank-two quotient is therefore a rational hyperbolic plane, and the form factors as

$$q_T(\xi) = c\,\xi(r)\xi(s)$$

with $c \in \mathbb{Q}^\times$ and nonzero $r, s \in S$. Evaluating at λ gives $\lambda(r)\lambda(s) = 0$. Injectivity (2) again yields a contradiction. Thus the rank is 3.

In either possible dimension, rank equals $\dim S$, so the range is all of S. □

4 The 2024–2026 frontier

Three recent developments are especially close to the remaining wall.

Matrix coefficients. Dasgupta and Kakde formulate the Matrix Coefficient Conjecture for matrices of complex or p-adic logarithms. In size two it contains the Four Exponentials problem rather than solving it: a zero determinant should become a zero coefficient after rational changes of basis. Their work clarifies the algebraic shape of the obstruction, but the required coefficient theorem remains conjectural in the present case [6].

Arithmetic holonomy. Calegari, Dimitrov, and Tang prove strong arithmetic-holonomy bounds and, among other applications, prove the $\mathbb{Q}$-linear independence of

$$1, \quad \log\left(1 + \frac{1}{m}\right), \quad \log\left(1 + \frac{1}{n}\right), \quad \log\left(1 + \frac{1}{m}\right)\log\left(1 + \frac{1}{n}\right)$$

when m/n is sufficiently close to 1 and $m \neq n$ [7]. The same authors' effective approximation theorem gives a uniform height bound for solutions of $|1 - A\gamma|_v \leq H(\gamma)^{-\varepsilon}$ in a fixed finitely generated multiplicative group [8]. These are real advances at weight two, but their proven product case lies on the special unit-gap locus $(m + 1)/m$.

Products of polylogarithms. Kawashima proves a new Padé criterion for linear independence of products of polylogarithms at distinct algebraic points [9]. Since

$$\mathrm{Li}_1(z) = -\log(1 - z),$$

this is the first theorem in the new literature whose conclusion directly contains mixed products of logarithms at many points. The rest of this report audits exactly how far its height hypothesis reaches toward (1).

5 Kawashima's weight-two defect

Let K be a number field, v a place, $m \geq 1$, and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m) \in (K^\times)^m$ with pairwise distinct coordinates. Put

$$\mathcal{D}_m = (m+1)^2 - 1 = m^2 + 2m$$

and

$$\mathcal{C}_m = \mathcal{D}_m \log 2 + 3\log(m+1) + 2 + 2\mathcal{D}_m. \tag{4}$$

For $\beta \in K$ satisfying $|\beta|_v > H_v(\boldsymbol{\alpha})$, Kawashima's weight-two defect is

$$\begin{aligned} \mathcal{V}_v(\boldsymbol{\alpha}, \beta) = (\mathcal{D}_m + 1)h_v(\beta) - h_v(\boldsymbol{\alpha}) \\ - \mathcal{D}_m \left(h(\beta) + \frac{1}{m}\sum_{i=1}^m h(\alpha_i) + h(\boldsymbol{\alpha}) \right) - \mathcal{C}_m. \end{aligned} \tag{5}$$

If this number is positive, then 1 together with all products of polylogarithms of total weight at most two at the points α_i/β is linearly independent over K. In particular, all numbers

$$1, \qquad \mathrm{Li}_1(\alpha_i/\beta), \qquad \mathrm{Li}_1(\alpha_i/\beta)\,\mathrm{Li}_1(\alpha_j/\beta)$$

are linearly independent over K.

The formula (5) looks representation-dependent: the same points $z_i = \alpha_i/\beta$ admit many common scalings. The next theorem removes that ambiguity by bounding the defect only in terms of the points themselves.

6 A representation-independent Padé obstruction

6.1 The projective defect theorem

Theorem 6.1 (Projective height-defect bound)**.** *Keep the notation above and put*

$$z_i = \frac{\alpha_i}{\beta}, \qquad a_v(\mathbf{z}) = -\log \max_i |z_i|_v, \qquad \overline{h}(\mathbf{z}) = \frac{1}{m}\sum_{i=1}^m h(z_i).$$

Then $a_v(\mathbf{z}) > 0$ *and*

$$\boxed{\mathcal{V}_v(\boldsymbol{\alpha}, \beta) \leq (\mathcal{D}_m + 1)a_v(\mathbf{z}) - \mathcal{D}_m\overline{h}(\mathbf{z}) - \mathcal{C}_m.} \tag{6}$$

The right-hand side is independent of the choice of common numerator and denominator.

Proof. Write

$$t = h_v(\beta) = \log|\beta|_v, \qquad A = h_v(\boldsymbol{\alpha}).$$

The strict size hypothesis implies $|\beta|_v > 1$. Since $\max_i |\alpha_i|_v = |\beta|_v \max_i |z_i|_v$, we have

$$A = \max\{0, t - a_v(\mathbf{z})\}. \tag{7}$$

Set

$$H = h(\beta), \qquad X = \frac{1}{m}\sum_i h(\alpha_i), \qquad Y = h(\boldsymbol{\alpha}), \qquad P = \overline{h}(\mathbf{z}).$$

The elementary height inequality $h(\alpha_i/\beta) \le h(\alpha_i) + h(\beta)$ gives $P \le X + H$. Also $H \ge t$ and $Y \ge A$. Therefore

$$H + X + Y \ge \max\{t, P\} + A. \tag{8}$$

Substituting (8) into (5) yields

$$\mathcal{V}_v \le (\mathcal{D}_m + 1)t - (\mathcal{D}_m + 1)\max\{0, t - a_v\} - \mathcal{D}_m \max\{t, P\} - \mathcal{C}_m. \tag{9}$$

Finally,

$$h(z_i) = h(z_i^{-1}) \ge h_v(z_i^{-1}) = -\log|z_i|_v,$$

so $P \ge a_v$. The piecewise-linear expression in (9) is then maximized for any t in the interval between a_v and P, where its value is $(\mathcal{D}_m + 1)a_v - \mathcal{D}_m P - \mathcal{C}_m$. This proves (6). □

PROVED.
Theorem 6.1 is the main new analytic theorem of this report. It applies at archimedean and nonarchimedean places, over arbitrary number fields, and optimizes the local scaling variable in the universal projective bound over every common scaling of $(\boldsymbol{\alpha}, \beta)$.

6.2 Arithmetic efficiency

Definition 6.2. For a tuple with $|z_i|_v < 1$, define its arithmetic efficiency at v by

$$\eta_v(\mathbf{z}) = \frac{a_v(\mathbf{z})}{\overline{h}(\mathbf{z})} \in (0, 1].$$

Corollary 6.3 (Efficiency and depth thresholds)**.** *If $\mathcal{V}_v(\boldsymbol{\alpha}, \beta) > 0$, then*

$$a_v(\mathbf{z}) > \mathcal{C}_m, \tag{10}$$

$$\eta_v(\mathbf{z}) > \frac{\mathcal{D}_m}{\mathcal{D}_m + 1} = 1 - \frac{1}{(m+1)^2}, \tag{11}$$

$$\overline{h}(\mathbf{z}) < \left(1 + \frac{1}{\mathcal{D}_m}\right) a_v(\mathbf{z}). \tag{12}$$

Proof. Since $\overline{h} \ge a_v$, positivity in (6) first gives $a_v > \mathcal{C}_m$. Dropping the positive constant from the same inequality gives $(\mathcal{D}_m + 1)a_v > \mathcal{D}_m \overline{h}$, which is (11); rearrangement gives (12). □

The numerical severity is visible immediately.

m	$\mathcal{D}_m$	$\mathcal{C}_m$	$e^{-\mathcal{C}_m}$	required η_v
1	3	12.158883	5.24×10^{-6}	> 0.750000
2	8	26.841014	2.20×10^{-12}	> 0.888889
3	15	46.556091	6.04×10^{-21}	> 0.937500
4	24	71.463846	9.20×10^{-32}	> 0.960000
5	35	101.635430	7.25×10^{-45}	> 0.972222
6	48	137.108795	2.85×10^{-60}	> 0.979592

Thus "make all points small" is not enough. Their local smallness must consume almost all of their global arithmetic height.

6.3 Exact scaling loss for integral presentations

Proposition 6.4 (Common scaling is strictly harmful)**.** *Let $K = \mathbb{Q}$, $v = \infty$, and let $1 \le \alpha_i < \beta$ be positive integers. Put $\alpha_{\max} = \max_i \alpha_i$. Then*

$$\mathcal{V}_\infty(\boldsymbol{\alpha}, \beta) = \log \beta - (\mathcal{D}_m + 1) \log \alpha_{\max} - \frac{\mathcal{D}_m}{m} \sum_i \log \alpha_i - \mathcal{C}_m. \tag{13}$$

For every integer $t \ge 1$,

$$\mathcal{V}_\infty(t\boldsymbol{\alpha}, t\beta) = \mathcal{V}_\infty(\boldsymbol{\alpha}, \beta) - 2\mathcal{D}_m \log t. \tag{14}$$

Proof. For positive integers, every finite-place local height vanishes, so $h(\beta) = h_\infty(\beta) = \log \beta$, $h(\alpha_i) = \log \alpha_i$, and $h(\boldsymbol{\alpha}) = h_\infty(\boldsymbol{\alpha}) = \log \alpha_{\max}$. Substitution gives (13); scaling all terms gives (14). □

This exact law explains why clearing a larger common denominator cannot rescue a failed specialization. It also corrects a tempting but false intuition: because the polylogarithm arguments are unchanged, one might expect the criterion to be projectively neutral. It is not; the arithmetic approximants remember the chosen affine presentation.

7 Natural encodings and why each misses the criterion

7.1 Direct prime-logarithm points

For every integer $q \ge 2$,

$$\log q = \mathrm{Li}_1\left(1 - \frac{1}{q}\right). \tag{15}$$

But $1 - 1/q \ge 1/2$. Hence any tuple containing a direct point has $a_\infty \le \log 2$, while Theorem 6.3 requires $a_\infty > \mathcal{C}_m \ge \mathcal{C}_1 > 12$. Therefore:

Corollary 7.1 (Direct points never qualify)**.** *No specialization of Kawashima's weight-two criterion containing even one point $1 - 1/q$ with $q \ge 2$ can have positive defect, regardless of the common representation or of the other points.*

This rules out the immediate choice

$$1/2, \quad 2/3, \quad 1 - 1/M, \quad 1 - 1/A$$

for the four logarithms in (1).

7.2 High-root splitting and degree dilution

A natural attempt to make the direct point small is

$$\log q = n\,\mathrm{Li}_1(1 - q^{-1/n}). \tag{16}$$

For $q = 2$ or 3, the polynomial $X^n - q$ is Eisenstein, so every number field containing $q^{1/n}$ has degree at least n. At the positive real place, Kawashima's normalized absolute value therefore gives

$$a_v = -\frac{1}{[K : \mathbb{Q}]} \log(1 - q^{-1/n}) \le \frac{1}{n} \log\left(1 + \frac{n}{\log q}\right) < 1.$$

Here we used $1 - e^{-y} \ge y/(1 + y)$ for $y > 0$. The depth condition $a_v > \mathcal{C}_m$ is impossible.

Corollary 7.2 (Radical splitting never qualifies)**.** *For either base logarithm* $\log 2$ *or* $\log 3$*, replacing it by arbitrary high-root identities of the form* (16) *cannot satisfy Kawashima's criterion at a real place. The apparent archimedean gain is more than canceled by the normalization factor* $[K:\mathbb{Q}]^{-1}$*.*

This is a useful warning for algebraic-point constructions: increasing the degree to make one chosen conjugate spectacularly close to 1 can reduce the favorable local height by exactly the same degree.

7.3 Telescoping by unit fractions

For integers $q \geq 2$ and $N \geq 1$,

$$\log q = \sum_{k=N+1}^{qN} \log \frac{k}{k-1} = \sum_{k=N+1}^{qN} \mathrm{Li}_1\left(\frac{1}{k}\right). \tag{17}$$

These points have perfect individual efficiency: for $z = 1/k$, $-\log z = h(z) = \log k$. Nevertheless the number of distinct points is $m = (q-1)N$, while the largest point is $1/(N+1)$. Thus

$$a_\infty = \log(N+1), \qquad \mathcal{C}_m > 2m^2.$$

Since $\log(N+1) < 2(q-1)^2N^2$ for every $q \geq 2, N \geq 1$, the necessary depth condition $a_\infty > \mathcal{C}_m$ fails before any finer height calculation is needed.

Corollary 7.3 (Dimension tax for ordinary telescoping)**.** *No ordinary telescoping decomposition* (17) *satisfies Kawashima's weight-two criterion.*

If one nevertheless clears all denominators with $L = \mathrm{lcm}(N+1, \ldots, qN)$, Theorem 6.4 makes the failure substantially worse. The exact integral defect is

$$-2\mathcal{D}_m \log L + (\mathcal{D}_m + 1)\log(N+1) + \frac{\mathcal{D}_m}{m} \log \frac{(qN)!}{N!} - \mathcal{C}_m,$$

which tends to $-\infty$ rapidly. The projective theorem shows that this is not a defect of that particular clearing; no other common scaling can reverse the sign.

8 A universal no-go theorem for rational Padé transfer

The preceding examples might suggest searching for a more ingenious rational family. The next theorem shows that this entire factor-by-factor strategy is impossible under the current criterion.

8.1 A valuation-lattice coefficient bound

Lemma 8.1 (Cramer bound for a rational multiplicative span)**.** *Let* $r_1, \ldots, r_m \in \mathbb{Q}_{>0}$*, let* $w_i = \boldsymbol{\nu}(r_i)$*, and suppose* $e_2 \in \mathrm{span}_{\mathbb{Q}}\{w_1, \ldots, w_m\}$*. If every coordinate of every* w_i *has absolute value at most* $E \geq 1$*, then there exist* $t \leq m$*, indices* $i_1, \ldots, i_t$*, a nonzero integer* d*, and integers* $c_1, \ldots, c_t$ *such that*

$$de_2 = \sum_{j=1}^{t} c_j w_{i_j} \tag{18}$$

and

$$|c_j| \le (t-1)^{(t-1)/2} E^{t-1}, \tag{19}$$

with the right-hand side interpreted as 1 *for* $t = 1$.

Proof. Choose t independent columns spanning e_2. Some $t \times t$ prime-row minor B is invertible. Such a minor necessarily contains the row indexed by 2: otherwise the restricted right-hand side is zero, so the uniquely determined coefficient vector is zero, contrary to $e_2 \neq 0$. Cramer's rule gives (18) with $d = \det B$ and the c_j equal to cofactors in the 2-row column. Hadamard's inequality on each $(t-1) \times (t-1)$ minor gives (19). □

8.2 No rational tuple can span $\log 2$

Theorem 8.2 (Rational-transfer impossibility). *Let* $z_1, \ldots, z_m \in \mathbb{Q} \cap (0,1)$ *be pairwise distinct. Suppose that they admit a specialization of Kawashima's weight-two criterion with positive defect at the real place. Put*

$$\ell_i = \mathrm{Li}_1(z_i) = -\log(1 - z_i).$$

Then

$$\log 2 \notin \mathrm{span}_{\mathbb{Q}}\{\ell_1, \ldots, \ell_m\}. \tag{20}$$

The same conclusion holds with 2 *replaced by any fixed rational number after changing only the harmless explicit constant in the proof.*

Proof. Set

$$a = -\log \max_i z_i, \qquad P = \frac{1}{m} \sum_i h(z_i), \qquad D = \mathcal{D}_m, \qquad C = \mathcal{C}_m.$$

By Theorem 6.3,

$$a > C, \qquad P < \left(1 + \frac{1}{D}\right) a. \tag{21}$$

Write $r_i = (1 - z_i)^{-1}$. If $z_i = u_i/v_i$ in lowest terms with $0 < u_i < v_i$, then $r_i = v_i/(v_i - u_i)$ is also in lowest terms and

$$h(r_i) = \log v_i = h(z_i). \tag{22}$$

Because all heights are nonnegative,

$$h(r_i) = h(z_i) \le mP < m\left(1 + \frac{1}{D}\right) a.$$

Every prime valuation of r_i is consequently bounded by

$$|v_p(r_i)| \le \frac{h(r_i)}{\log 2} < 3ma =: E. \tag{23}$$

Also $a > C > \log 2$, so $z_i \le e^{-a} < 1/2$ and

$$0 < \ell_i = -\log(1 - z_i) \le \frac{z_i}{1 - z_i} \le 2e^{-a}. \tag{24}$$

Assume for contradiction that $\log 2$ belongs to the rational span of the ℓ_i. Exponentiating after clearing denominators shows that e_2 belongs to the rational span of the valuation vectors

$w_i = \boldsymbol{\nu}(r_i)$. Apply Theorem 8.1. Taking logarithms in (18) and using (24) gives

$$\log 2 \le |d| \log 2 = \left| \sum_{j=1}^{t} c_j \ell_{i_j} \right| \tag{25}$$
$$\le 2t(t-1)^{(t-1)/2}(3ma)^{t-1} e^{-a} \tag{26}$$
$$\le 2m\, m^{(m-1)/2}(3ma)^{m-1} e^{-a}. \tag{27}$$

It remains only to check that the last expression is smaller than $\log 2$ whenever $a > C$.

The function $(3ma)^{m-1}e^{-a}$ is decreasing for $a > m-1$, and $C > 2m^2 > m-1$. Furthermore

$$2m^2 + 4m < C < 4(m+1)^2. \tag{28}$$

The upper bound follows by writing $y = m+1 \ge 2$ and observing that $(2-\log 2)y^2 + \log 2 - 3\log y > 0$; the lower bound is immediate from $C = (2+\log 2)(m^2+2m) + 3\log(m+1) + 2$. At $a = C$, the logarithm of the right-hand side of (27), divided by $\log 2$, is at most

$$\log(3m) + (m-1)(\log 48 + 3.5\log m) - 2m^2 - 4m.$$

For $m = 1$ the inequality is immediate. For real $m \ge 2$, the negative of this expression is positive at $m = 2$ and has positive derivative; its derivative itself is increasing because $4 - 3.5/m - 2.5/m^2 > 0$. Hence the expression is negative for every integer $m \ge 2$. This contradicts (27). □

PROVED.
The theorem is universal in m. It does not merely say that the obvious representations of $\log 2$ fail; it says that *every rational tuple satisfying Kawashima's own height hypothesis* is too arithmetically close to 1 for its valuation lattice to recover the fixed prime direction e_2.

8.3 The full formal quadratic transfer also fails

Let $W = \operatorname{span}_{\mathbb{Q}}\{\boldsymbol{\nu}((1-z_i)^{-1}) : 1 \le i \le m\}$. Any factor-by-factor rational logarithm identity places the resulting quadratic forms in $\operatorname{Sym}^2(W)$. The tensor-rank audit from Theorem 3.1 now gives a stronger corollary.

Corollary 8.3 (No formal rational quadratic transfer)**.** *Assume a nonintegral Alaoglu–Erdős counterexample and let T be its prime-valuation tensor* (3)*. There is no rational tuple* $z_1, \dots, z_m \in (0,1)$ *such that*

(i) *a specialization of Kawashima's weight-two criterion at these points has positive defect;*
(ii) $T \in \operatorname{Sym}^2(W)$ *for* $W = \operatorname{span}_{\mathbb{Q}}\{\boldsymbol{\nu}((1-z_i)^{-1})\}$.

In particular, one cannot settle (1) *by expressing its four logarithms through rational* Li_1 *identities and then invoking Kawashima's product independence.*

Proof. Every tensor in $\operatorname{Sym}^2(W)$ has contraction range contained in W. By Theorem 3.1, the range of T is $S = \operatorname{span}\{u, v, e_2, e_3\}$, so $e_2 \in W$. Applying λ gives $\log 2$ in the rational span of the numbers $\log(1-z_i)^{-1} = \mathrm{Li}_1(z_i)$, contradicting Theorem 8.2. □

The corollary leaves a narrow but genuine escape hatch: one could use algebraic points, unbounded number-field degree, and cross-embedding or norm identities that do not reduce to a rational prime-valuation tensor inside one fixed field. Sections below show why the obvious versions of that escape hatch also encounter strong barriers.

9 Fixed multiplicative groups: effective exceptional finiteness

The projective defect bound combines cleanly with the effective approximation theorem of Calegari–Dimitrov–Tang.

Theorem 9.1 (Fixed-group Padé tuples are effectively finite)**.** *Fix a number field K, a place v, a finitely generated subgroup $\Gamma < K^\times$, and an integer $m \geq 1$. Consider pairwise distinct $\gamma_1, \dots, \gamma_m \in \Gamma$ such that*

$$z_i = 1 - \gamma_i, \qquad |z_i|_v < 1.$$

There are only finitely many such tuples for which the points z_i admit a specialization of Kawashima's weight-two criterion with positive defect. Their heights are effectively bounded in terms of (K, v, Γ, m).

Proof. Let $a = a_v(\mathbf{z})$ and $P = \overline{h}(\mathbf{z})$. Positivity and (12) imply, for every i,

$$h(z_i) \leq mP < \frac{m(\mathcal{D}_m + 1)}{\mathcal{D}_m} a.$$

Hence, with

$$\varepsilon_m = \frac{\mathcal{D}_m}{2m(\mathcal{D}_m + 1)},$$

we have $a > 2\varepsilon_m h(z_i)$. The standard height inequalities give $|h(\gamma_i) - h(z_i)| \leq \log 2$. If $h(\gamma_i) \leq 2 \log 2$, it is already bounded. Otherwise

$$-\log |1 - \gamma_i|_v = -\log |z_i|_v \geq a > \varepsilon_m h(\gamma_i),$$

so

$$|1 - \gamma_i|_v < H(\gamma_i)^{-\varepsilon_m}.$$

The effective approximation theorem of [8], applied with $A = 1$, bounds $h(\gamma_i)$ effectively. Northcott finiteness in the fixed field K then leaves only finitely many tuples. □

Remark 9.2. The theorem does not prove Alaoglu–Erdős. Under a hypothetical counterexample the group of S-units generated by the finitely many primes dividing $6MA$ is fixed, but the effective exceptional set may be nonempty and its defining set S is itself unknown. What the theorem does prove is that no asymptotic sequence inside a fixed S-unit group can drive Kawashima's defect positive. Any such proof would have to come from a finite exceptional configuration or from fields/groups whose arithmetic complexity changes with the construction.

10 The near-unit unimodular normal form

10.1 Exact construction

Let

$$\rho = \frac{\log 3}{\log 2}$$

and let p_n/q_n be its continued-fraction convergents. Define

$$u_n = 2^{p_n} 3^{-q_n}, \qquad U_n = M^{p_n} A^{-q_n} = u_n^x.$$

Let $\varepsilon_n \in \{\pm 1\}$ orient the first unit above 1:

$$r_n = u_n^{\varepsilon_n} > 1, \qquad R_n = U_n^{\varepsilon_n} = r_n^x > 1.$$

Consecutive convergents satisfy $p_n q_{n+1} - p_{n+1} q_n = \pm 1$. Therefore (r_n, r_{n+1}) is a multiplicative basis of $\langle 2, 3\rangle$ and (R_n, R_{n+1}) is a multiplicative basis of $\langle M, A\rangle$. Moreover

$$\det \begin{pmatrix} \log r_n & \log r_{n+1} \\ \log R_n & \log R_{n+1} \end{pmatrix} = 0 \tag{29}$$

and all four units tend to 1.

The usual convergent inequalities give

$$\frac{\log 2}{q_n + q_{n+1}} < |\log u_n| < \frac{\log 2}{q_{n+1}}. \tag{30}$$

Thus a counterexample is equivalent to infinitely many exact weight-two determinant zeros in an arbitrarily small neighborhood of $(1, 1, 1, 1)$.

10.2 Why mere proximity is not enough

For a positive rational unit $r > 1$, put

$$z(r) = 1 - r^{-1}, \qquad \eta(r) = \frac{-\log z(r)}{h(z(r))}.$$

If $r = N/D$ is reduced, then $z(r) = (N - D)/N$ is reduced and $h(z(r)) = \log N = h(r)$. The ratios $(m+1)/m$ appearing in the product theorem of Calegari–Dimitrov–Tang have numerator gap 1 and therefore

$$\eta\left(\frac{m+1}{m}\right) = 1.$$

They lie on the maximal-efficiency boundary.

The convergent units are entirely different. A lower bound for a nonzero linear form in $\log 2$ and $\log 3$ gives effective constants $c, C > 0$ such that

$$|p \log 2 - q \log 3| > c \max\{|p|, |q|\}^{-C}.$$

Together with (30), this implies

$$-\log z(r_n) = O(\log q_n), \qquad h(z(r_n)) \asymp q_n,$$

so

$$\eta(r_n) \longrightarrow 0. \tag{31}$$

The same conclusion holds for R_n. Indeed $\log R_n = x \log r_n$, while the valuation vectors $p_n \boldsymbol{\nu}(M) - q_n \boldsymbol{\nu}(A)$ have norm $\asymp q_n$ because M and A are multiplicatively independent.

PROVED.
The natural microlocal form of a counterexample lives in the *low-efficiency* regime $\eta \to 0$. Kawashima's present criterion requires average efficiency $> 1 - 1/(m+1)^2$, and the proved arithmetic-holonomy product theorem treats the unit-gap locus $\eta = 1$. The missing theorem must therefore cross an arithmetic-efficiency gap, not merely a geometric "points close to one" gap.

10.3 A numerical sample

n	p_n	q_n	$\lvert p_n \log 2 - q_n \log 3\rvert$	height scale
0	1	1	4.0547×10^{-1}	1
1	2	1	2.8768×10^{-1}	2
2	3	2	1.1778×10^{-1}	3
3	8	5	5.2116×10^{-2}	8
4	19	12	1.3551×10^{-2}	19
5	65	41	1.1463×10^{-2}	65
6	84	53	2.0881×10^{-3}	84
7	485	306	1.0222×10^{-3}	485
8	1054	665	4.3654×10^{-5}	1054
9	24727	15601	1.8195×10^{-5}	24727

The relevant decay is polynomial in the exponent size, while high-efficiency Padé criteria need a gap essentially as small as the reciprocal arithmetic denominator.

11 Why the remaining nearby tools do not transfer

11.1 p-adic specialization

Theorem 6.1 applies just as well at a finite place, and in some examples a p-adic local contribution can be much more favorable. However Kawashima's conclusion at a finite place is linear independence of p-adic polylogarithm values in K_v. The real relation (1) does not automatically become a relation between p-adic logarithms. A comparison principle strong enough to transfer that relation would itself be a new period or motivic theorem at approximately the same transcendence level as the original problem. Thus choosing a finite place changes the values being proved independent; it does not simply improve the height estimate for the real values.

11.2 Powers of one logarithm

Kawashima's appendix constructs new Padé-type approximants for

$$\log(1 - 1/z), \ldots, \log^m(1 - 1/z)$$

and proves the relevant functional determinant is nonzero. This is valuable structural input, but (1) is a cross-product of logarithms at different algebraic arguments. Moreover the paper explicitly leaves the Poincaré-type recurrence asymptotics needed for sharp numerical independence measures to future work. A one-point power theorem cannot collapse the multiplicative rank of 2 and 3 to one.

11.3 The proved holonomy product theorem

The theorem of Calegari–Dimitrov–Tang proves a genuine mixed product result, but specifically for

$$\log\left(1 + \frac{1}{m}\right) \log\left(1 + \frac{1}{n}\right)$$

with m/n close to 1. A convergent unit $2^p/3^q$ generally has numerator gap $|2^p - 3^q| \gg 1$, so it is not of the form $(m+1)/m$. Pairwise irrationality of individual products also does not imply joint linear independence of all products arising after a long telescoping decomposition. The theorem reaches weight two, but on a high-efficiency locus disjoint from the canonical counterexample normal form.

12 What a successful Padé continuation would have to change

The no-go theorems above identify three requirements that are absent from the current criterion.

12.1 Direct determinant approximation rather than monomial independence

Kawashima proves the independence of a very large family: all polylogarithm monomials of total weight at most two. For m points this family has size $(m+1)^2$, and the constant $\mathcal{C}_m \asymp (2+\log 2)m^2$ records that dimensional cost. Alaoglu–Erdős needs only one specific determinant. A tailored construction should approximate

$$\Delta = \log M \log 3 - \log A \log 2$$

directly instead of paying for every monomial.

12.2 Separate local denominators

The common parameter β forces all points through one affine denominator and charges the heights of both the coordinates and their vector. A plausible replacement is a genuinely mixed-denominator Hermite–Padé system with one local denominator for each point, followed by adelic cancellation *before* a common denominator is cleared. Merely rewriting the same points with a larger common denominator is impossible by (14).

12.3 Valuation-lattice cancellation

The rational-transfer theorem shows that individual high-efficiency logarithms are too small to recover a fixed prime direction with the polynomially bounded coefficients supplied by a valuation lattice. A successful construction must therefore exploit cancellation at the level of the *four-log determinant*: prime factors appearing in one Padé row must cancel against prime factors in another row. This is precisely the type of cross-row gain that the ordinary one-row height defect discards.

The following statement isolates a sufficient target without pretending that it is already proved.

Conjecture 12.1 (Mixed-denominator determinant criterion)**.** *For positive rationals r_1, r_2, s_1, s_2 such that each pair is multiplicatively independent, there is a Padé construction for*

$$\log r_1 \log s_2 - \log r_2 \log s_1$$

whose denominator estimate is governed by the determinantal divisors of the two prime-valuation matrices, including their saturation indices, rather than by the sum of the four individual affine heights. The construction proves that the displayed determinant is nonzero unless a rational row/column change produces a zero logarithmic coefficient.

Proposition 12.2. *Theorem 12.1 implies the Alaoglu–Erdős conjecture.*

Proof. Assume a counterexample. Use consecutive convergents as in Equation (29). The pairs (r_n, r_{n+1}) and (R_n, R_{n+1}) are unimodular changes of the respective rank-two valuation-lattice generators, while their logarithmic determinant is zero. The exceptional alternative in Theorem 12.1 would give a zero logarithmic coefficient after rational basis changes. By injectivity of the rational logarithm map, that coefficient is zero exactly when one of the corresponding multiplicative basis elements is 1, impossible. Hence the determinant must be nonzero, contradicting (29). □

The conjectural criterion is deliberately narrower than Four Exponentials in arbitrary algebraic settings: it asks only for positive rationals and rank-two prime-valuation lattices. It is also deliberately stronger than the existing Padé theorems in the one place where the present problem needs strength: low-efficiency, cross-row denominator cancellation.

13 Directed-rounding extension through 2^{29}

13.1 Verifier contract

Let

$$\rho = \frac{\log 3}{\log 2}.$$

For each positive integer m that is not a power of two, the verifier seeks an integer n with

$$n < m^\rho < n + 1.$$

It never evaluates m^ρ. At 192-bit MPFR precision it computes directed enclosures of the needed logarithms and proves the two strict inequalities through positive lower bounds for

$$\delta_- = \log(m)\log 3 - \log(n)\log 2, \tag{32}$$
$$\delta_+ = \log(n+1)\log 2 - \log(m)\log 3. \tag{33}$$

A long-double calculation proposes n, but the proposal has no logical role: acceptance requires both MPFR lower bounds to be strictly positive. A locator error can therefore create a reported failure, not a false certificate.

The exact C source was extracted from the supplied report without modification. Its SHA-256 digest is

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

This identity matches the source digest already printed in the supplied report.

13.2 New range and aggregate result

The new audit covers two consecutive bit ranges,

$$[2^{27}, 2^{29}) = [134{,}217{,}728, 536{,}870{,}912),$$

partitioned into 96 chunks of width 2^{22}. The powers 2^{27} and 2^{28} are skipped. Consequently the exact number of tested inputs is

$$(2^{29} - 2^{27}) - 2 = 402{,}653{,}182.$$

The first 32 chunks cover $[2^{27}, 2^{28})$; the next 64 cover the entire range $[2^{28}, 2^{29})$.

Range	tested nonpowers	failures	summed chunk time (s)
$[2^{27}, 2^{28})$	134,217,727	0	395.843
$[2^{28}, 2^{29})$	268,435,455	0	792.962
Combined new audit	402,653,182	0	1,188.805

Across both new ranges, the smallest certified lower-side margin was

$$\delta_- > 4.9141462461940518 \times 10^{-24} \quad \text{at} \quad (m, n) = (388{,}909{,}543, 41{,}170{,}668{,}399{,}174), \tag{34}$$

and the smallest certified upper-side margin was

$$\delta_+ > 5.2103670350428438 \times 10^{-23} \quad \text{at} \quad (m, n) = (501{,}373{,}708, 61{,}578{,}600{,}753{,}798). \tag{35}$$

Both extremal records were replayed with one thread at 1024-bit MPFR precision. The corresponding directed lower bounds were unchanged to the displayed 17 digits. As a separate implementation check, a 320-decimal-digit evaluation gave

$$\delta_- = 4.914146246194051848982433252362132380596\ldots \times 10^{-24}$$

for (34), and

$$\delta_+ = 5.210367035042843821555189929112177636068\ldots \times 10^{-23}$$

for (35). This decimal replay is corroborative only; the certificate is the positive directed MPFR lower bound.

The executable used GCC 14.2.0, MPFR 4.2.2, 192-bit precision, and five OpenMP threads for each chunk. The source and executable SHA-256 digests were, respectively,

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`,

`85f2acca22ec3ea4bf783a2e13356a1e11d9faff3b95a4d1c79a7cced69deceb`.

The concatenated logs for the two blocks had SHA-256 digests

$[2^{27}, 2^{28})$: `3865fb3d99d3a6df4ab4ac101942781dea1065f38414462fb2e25b0ba4ef16ce`,

$[2^{28}, 2^{29})$: `6899851a9d78bca4e7527011c10089aaede12f1f93a9584efdcaa0d5d56cebe2`.

The two 1024-bit replay logs had digests

`333b015a3ac541f34a4c14197dc67d03d5211f7bc0828abd346695790a174555`,

`de0eac89dcc88c63f080e8714fa5e93f0485beb8e913e87ab705316b80be619b`,

and the independent 320-digit replay file had digest

`791d592dc7140674e8300c2ae65db6ef6be9f539b1f92a05a8b6f5f4c37be223`.

COMPUTATION.
The result is a software-level directed-rounding certificate, not a proof-assistant kernel certificate. It depends on the stated MPFR implementation, compiler, ABI declarations, and executed binary. Those trust assumptions are materially stronger than ordinary floating-point sampling but materially weaker than a formal proof.

13.3 Consequence for the exponent

Proposition 13.1 (Finite exclusion through 29)**.** *At the trust level of the directed-rounding computation, any nonintegral counterexample satisfies* $x > 29$*.*

Proof. Suppose $x < 29$ and $M = 2^x \in \mathbb{N}$. Then $M < 2^{29}$. If M is a power of two, injectivity of the real exponential gives $x \in \mathbb{Z}$. Otherwise the inherited scan below 2^{27} together with the two new ranges proves that $M^\rho = 3^x$ lies strictly between consecutive integers, contradicting the hypothesis that 3^x is an integer. The endpoint $x = 29$ is integral. □

14 Synthesis

The post-report Padé literature gets closer to the correct transcendence weight than any of the classical one-logarithm tools, but the new analysis identifies an exact mismatch.

- **a.** A counterexample is naturally converted by continued fractions into determinant zeros among rational units arbitrarily close to 1.
- **b.** Those units have arithmetic efficiency tending to 0: their gaps are only polynomially small in the exponent size, while their rational heights are exponential in that size.
- **c.** Kawashima's current theorem requires average efficiency $> 1 - 1/(m+1)^2$ and a local depth exceeding a quadratic dimension constant.
- **d.** The proven arithmetic-holonomy product theorem occupies the extreme unit-gap locus of efficiency 1.
- **e.** Rational algebraic transfer cannot bridge the gap: the valuation lattice bounds the required coefficients polynomially, while all qualifying logarithms are exponentially small. This is the content of Theorem 8.2.

The remaining route is therefore not "find a cleverer common denominator." It is to build a weight-two approximant that sees the prime-valuation determinant of the *whole* expression and cancels denominators across rows before individual heights are charged. That target is technically concrete, but it is not presently a theorem.

15 Final status

OPEN / CONDITIONAL.
No unconditional proof of

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}$$

is contained in this report. The exact unresolved statement remains the nonvanishing of the weight-two logarithmic determinant (1), equivalently the corresponding restricted matrix-coefficient assertion.

The rigorous advances are nonetheless substantive:

- a new projective upper bound for the 2026 polylogarithm-product criterion;
- a quantitative arithmetic-efficiency threshold with sharp dependence on the number of points;
- complete no-go results for direct points, radical splitting, ordinary telescoping, and common scaling;
- a universal rational-transfer impossibility theorem, plus a tensor-level corollary for the formal Alaoglu–Erdős quadratic relation;
- effective finiteness of every fixed-group Padé attempt;
- a precise diagnosis of the high-efficiency/low-efficiency gap between current theorems and the canonical counterexample normal form;
- an independently rerun directed-rounding exclusion through 2^{29}.

These results eliminate a broad family of plausible but ultimately nonviable proof strategies and leave a narrower, technically explicit research target: direct determinant Padé approximation with cross-row valuation cancellation.

A Elementary inequalities used in the rational-transfer theorem

For completeness, this appendix records the two estimates compressed in Theorem 8.2.

Let $D = m^2 + 2m$ and
$$C = (2 + \log 2)D + 3\log(m + 1) + 2.$$
Then $C > 2m^2 + 4m$ is immediate. For the upper bound, put $y = m + 1 \geq 2$:
$$\begin{aligned} 4(m+1)^2 - C &= 4y^2 - (2 + \log 2)(y^2 - 1) - 3\log y - 2 \\ &= (2 - \log 2)y^2 + \log 2 - 3\log y. \end{aligned}$$
The last expression is positive at $y = 2$, and its derivative $2(2 - \log 2)y - 3/y$ is positive for $y \geq 2$. Hence
$$2m^2 + 4m < C < 4(m + 1)^2.$$
For $m \geq 2$, define
$$G(m) = 2m^2 + 4m - \log(3m) - (m - 1)(\log 48 + 3.5\log m).$$
Directly, $G(2) > 7.9$. Moreover
$$G'(m) = 4m + \frac{1}{2} + \frac{2.5}{m} - \log 48 - 3.5\log m > 0$$
for $m \geq 2$, because its derivative $4 - 3.5/m - 2.5/m^2$ is already positive at $m = 2$. Thus $G(m) > 0$ for all integers $m \geq 2$. This is the last numerical inequality needed to make (27) impossible.

B Scan chunk manifest

The table records every chunk. Each interval has width 2^{22}; "start" is its inclusive lower endpoint. A record d (m) means that the chunk minimum was the directed lower bound d

attained at that value of m. The corresponding integer n is retained in the hashed raw log; the two global n values are displayed in (34)–(35).

Table 1: Directed-rounding chunk manifest for $[2^{27}, 2^{29})$.

ID	start	tested	minimum δ_- (m)	minimum δ_+ (m)	seconds
27.00	134,217,728	4,194,303	$4.284e-20$ (136,640,874)	$3.389e-20$ (135,357,389)	11.825
27.01	138,412,032	4,194,304	$2.815e-20$ (138,540,675)	$2.847e-21$ (142,153,616)	12.013
27.02	142,606,336	4,194,304	$1.061e-20$ (142,695,401)	$6.273e-22$ (146,333,091)	12.881
27.03	146,800,640	4,194,304	$3.593e-21$ (148,583,161)	$1.143e-20$ (148,016,970)	11.572
27.04	150,994,944	4,194,304	$3.782e-21$ (153,415,953)	$4.828e-20$ (154,129,681)	13.257
27.05	155,189,248	4,194,304	$1.677e-20$ (159,254,759)	$1.606e-20$ (158,382,018)	12.778
27.06	159,383,552	4,194,304	$3.469e-20$ (162,200,015)	$2.021e-20$ (160,095,403)	11.821
27.07	163,577,856	4,194,304	$3.847e-21$ (166,033,450)	$7.569e-21$ (166,969,115)	12.110
27.08	167,772,160	4,194,304	$8.665e-21$ (171,251,435)	$1.465e-20$ (170,443,795)	11.822
27.09	171,966,464	4,194,304	$4.262e-20$ (173,816,254)	$1.433e-20$ (175,647,581)	11.661
27.10	176,160,768	4,194,304	$1.876e-20$ (178,598,559)	$4.477e-21$ (179,458,795)	13.244
27.11	180,355,072	4,194,304	$5.679e-20$ (180,765,336)	$5.445e-22$ (183,569,533)	12.511
27.12	184,549,376	4,194,304	$3.085e-22$ (187,004,237)	$7.872e-21$ (188,301,745)	12.227
27.13	188,743,680	4,194,304	$1.102e-20$ (192,461,540)	$1.798e-20$ (191,799,137)	12.614
27.14	192,937,984	4,194,304	$1.638e-21$ (195,368,525)	$2.173e-20$ (194,355,394)	11.846
27.15	197,132,288	4,194,304	$8.763e-21$ (197,987,777)	$1.177e-20$ (198,330,250)	12.451
27.16	201,326,592	4,194,304	$9.306e-22$ (202,612,057)	$9.219e-21$ (203,389,040)	13.492
27.17	205,520,896	4,194,304	$2.745e-21$ (206,738,514)	$1.370e-21$ (206,597,929)	11.626
27.18	209,715,200	4,194,304	$1.383e-20$ (213,551,311)	$8.366e-21$ (213,290,074)	13.091
27.19	213,909,504	4,194,304	$4.101e-22$ (214,684,191)	$1.679e-21$ (217,426,809)	12.462
27.20	218,103,808	4,194,304	$3.638e-21$ (218,691,392)	$2.873e-21$ (218,721,921)	12.370
27.21	222,298,112	4,194,304	$1.825e-20$ (224,977,296)	$5.053e-21$ (226,024,581)	12.121
27.22	226,492,416	4,194,304	$1.646e-20$ (226,553,247)	$2.228e-20$ (230,684,216)	13.078
27.23	230,686,720	4,194,304	$1.953e-22$ (233,957,087)	$6.347e-21$ (231,792,276)	12.023
27.24	234,881,024	4,194,304	$3.812e-21$ (235,808,239)	$3.750e-21$ (236,568,448)	12.491
27.25	239,075,328	4,194,304	$6.046e-21$ (239,456,283)	$9.724e-21$ (239,580,935)	13.390
27.26	243,269,632	4,194,304	$3.971e-21$ (244,524,864)	$5.006e-21$ (245,994,981)	11.525
27.27	247,463,936	4,194,304	$1.077e-20$ (249,872,818)	$8.052e-21$ (249,272,020)	12.868
27.28	251,658,240	4,194,304	$1.929e-20$ (251,728,391)	$4.421e-21$ (254,182,804)	12.215
27.29	255,852,544	4,194,304	$6.990e-21$ (256,676,151)	$8.307e-22$ (259,598,690)	11.862
27.30	260,046,848	4,194,304	$4.051e-21$ (263,613,699)	$9.409e-22$ (261,264,827)	12.792
27.31	264,241,152	4,194,304	$1.374e-21$ (266,715,185)	$4.150e-21$ (264,885,749)	11.805
28.00	268,435,456	4,194,303	$1.438e-20$ (270,769,256)	$1.276e-20$ (269,492,939)	12.071
28.01	272,629,760	4,194,304	$1.294e-20$ (274,601,332)	$1.329e-20$ (276,729,727)	11.772
28.02	276,824,064	4,194,304	$1.336e-21$ (278,937,929)	$1.473e-20$ (279,209,135)	13.839
28.03	281,018,368	4,194,304	$6.220e-21$ (283,347,824)	$2.847e-21$ (284,307,232)	12.520
28.04	285,212,672	4,194,304	$3.573e-21$ (285,482,152)	$1.521e-20$ (288,577,614)	12.190
28.05	289,406,976	4,194,304	$8.508e-21$ (290,399,029)	$6.273e-22$ (292,666,182)	12.526
28.06	293,601,280	4,194,304	$3.593e-21$ (297,166,322)	$3.235e-21$ (297,393,168)	12.063
28.07	297,795,584	4,194,304	$1.775e-21$ (301,090,986)	$3.276e-20$ (298,685,330)	12.413
28.08	301,989,888	4,194,304	$1.725e-21$ (304,348,427)	$3.239e-21$ (303,439,436)	13.260
28.09	306,184,192	4,194,304	$5.191e-22$ (306,228,765)	$2.879e-20$ (309,957,834)	11.720
28.10	310,378,496	4,194,304	$3.887e-21$ (313,261,071)	$5.204e-21$ (313,114,243)	12.893
28.11	314,572,800	4,194,304	$1.048e-20$ (316,959,535)	$3.653e-22$ (314,764,391)	12.743
28.12	318,767,104	4,194,304	$1.666e-20$ (321,040,811)	$2.032e-21$ (320,803,384)	12.558
28.13	322,961,408	4,194,304	$1.378e-20$ (324,726,584)	$2.016e-20$ (325,862,257)	12.591
28.14	327,155,712	4,194,304	$9.296e-21$ (330,041,528)	$1.951e-21$ (327,923,577)	12.732
28.15	331,350,016	4,194,304	$3.847e-21$ (332,066,900)	$6.512e-21$ (334,016,730)	12.549
28.16	335,544,320	4,194,304	$1.113e-21$ (338,945,756)	$2.743e-21$ (336,494,503)	12.405
28.17	339,738,624	4,194,304	$7.650e-21$ (343,687,066)	$1.989e-21$ (343,467,223)	12.313
28.18	343,932,928	4,194,304	$2.000e-21$ (347,267,147)	$2.772e-21$ (346,761,393)	12.206
28.19	348,127,232	4,194,304	$1.437e-22$ (349,972,684)	$1.433e-20$ (351,295,162)	13.603
28.20	352,321,536	4,194,304	$5.143e-22$ (353,661,837)	$2.833e-21$ (353,990,351)	11.765
28.21	356,515,840	4,194,304	$3.081e-21$ (356,895,426)	$5.735e-22$ (357,765,549)	11.813
28.22	360,710,144	4,194,304	$8.141e-22$ (364,737,335)	$1.316e-22$ (362,062,452)	13.552
28.23	364,904,448	4,194,304	$9.240e-22$ (365,075,693)	$3.223e-22$ (365,778,901)	11.562
28.24	369,098,752	4,194,304	$7.157e-22$ (371,893,339)	$8.105e-22$ (371,246,341)	11.810
28.25	373,293,056	4,194,304	$3.085e-22$ (374,008,474)	$2.529e-21$ (376,055,882)	13.831
28.26	377,487,360	4,194,304	$8.880e-21$ (378,129,221)	$6.429e-21$ (381,433,624)	12.044
28.27	381,681,664	4,194,304	$8.273e-22$ (384,046,343)	$1.056e-20$ (382,597,108)	12.525
28.28	385,875,968	4,194,304	$4.914e-24$ (388,909,543)	$8.857e-21$ (386,125,278)	12.275
28.29	390,070,272	4,194,304	$1.638e-21$ (390,737,050)	$8.114e-22$ (391,647,767)	11.700
28.30	394,264,576	4,194,304	$2.526e-21$ (395,220,414)	$5.912e-21$ (398,186,911)	12.558

continued on next page

Table 1 continued

ID	start	tested	minimum δ_- (m)	minimum δ_+ (m)	seconds
28.31	398,458,880	4,194,304	$7.885e-24$ (400,544,529)	$1.694e-20$ (402,139,190)	12.558
28.32	402,653,184	4,194,304	$9.306e-22$ (405,224,114)	$4.711e-22$ (405,024,519)	12.028
28.33	406,847,488	4,194,304	$9.700e-22$ (407,982,235)	$2.819e-21$ (407,281,054)	12.332
28.34	411,041,792	4,194,304	$2.745e-21$ (413,477,028)	$1.293e-21$ (412,276,830)	11.854
28.35	415,236,096	4,194,304	$5.219e-21$ (416,468,676)	$7.693e-22$ (418,544,766)	11.578
28.36	419,430,400	4,194,304	$1.954e-21$ (422,398,112)	$3.436e-21$ (423,295,624)	12.982
28.37	423,624,704	4,194,304	$1.289e-21$ (423,710,706)	$2.513e-22$ (426,529,906)	12.895
28.38	427,819,008	4,194,304	$4.101e-22$ (429,368,382)	$4.883e-22$ (429,282,395)	13.442
28.39	432,013,312	4,194,304	$5.984e-21$ (432,505,720)	$1.679e-21$ (434,853,618)	12.570
28.40	436,207,616	4,194,304	$1.685e-21$ (436,314,750)	$2.873e-21$ (437,443,842)	11.926
28.41	440,401,920	4,194,304	$1.275e-21$ (442,841,765)	$5.083e-21$ (442,062,629)	11.858
28.42	444,596,224	4,194,304	$4.070e-21$ (447,235,367)	$7.158e-21$ (447,720,068)	13.405
28.43	448,790,528	4,194,304	$1.569e-21$ (452,029,583)	$2.822e-22$ (450,505,498)	12.393
28.44	452,984,832	4,194,304	$4.410e-21$ (455,437,571)	$1.014e-21$ (454,240,771)	11.531
28.45	457,179,136	4,194,304	$2.793e-21$ (459,828,193)	$8.322e-21$ (461,365,410)	12.483
28.46	461,373,440	4,194,304	$1.747e-21$ (463,276,633)	$6.347e-21$ (463,584,552)	11.568
28.47	465,567,744	4,194,304	$1.953e-22$ (467,914,174)	$7.459e-22$ (466,121,593)	11.721
28.48	469,762,048	4,194,304	$3.067e-21$ (471,111,503)	$3.750e-21$ (473,136,896)	13.749
28.49	473,956,352	4,194,304	$3.276e-21$ (477,377,170)	$2.199e-22$ (474,560,743)	11.791
28.50	478,150,656	4,194,304	$5.934e-21$ (480,943,527)	$6.086e-21$ (478,837,280)	11.811
28.51	482,344,960	4,194,304	$4.365e-22$ (485,602,436)	$4.656e-22$ (483,731,011)	12.214
28.52	486,539,264	4,194,304	$5.845e-22$ (488,578,708)	$2.469e-22$ (490,595,136)	12.000
28.53	490,733,568	4,194,304	$6.973e-21$ (494,115,822)	$1.425e-21$ (491,328,133)	11.615
28.54	494,927,872	4,194,304	$4.455e-21$ (495,724,767)	$1.792e-21$ (496,559,778)	14.097
28.55	499,122,176	4,194,304	$5.780e-22$ (499,782,534)	$5.210e-23$ (501,373,708)	11.701
28.56	503,316,480	4,194,304	$4.533e-21$ (504,996,574)	$9.370e-22$ (504,517,808)	11.720
28.57	507,510,784	4,194,304	$1.140e-21$ (507,876,391)	$1.183e-21$ (509,911,917)	12.174
28.58	511,705,088	4,194,304	$6.990e-21$ (513,352,302)	$4.274e-22$ (514,216,147)	12.008
28.59	515,899,392	4,194,304	$1.115e-21$ (516,239,075)	$8.307e-22$ (519,197,380)	12.192
28.60	520,093,696	4,194,304	$6.023e-22$ (521,145,596)	$9.409e-22$ (522,529,654)	14.640
28.61	524,288,000	4,194,304	$1.326e-21$ (528,412,351)	$1.972e-21$ (526,958,976)	12.038
28.62	528,482,304	4,194,304	$4.360e-21$ (528,890,012)	$9.307e-22$ (530,509,954)	11.560
28.63	532,676,608	4,194,304	$1.374e-21$ (533,430,370)	$4.753e-21$ (535,448,067)	12.130

C Reproducibility commands

The source was compiled and the new range was scanned with commands equivalent to:

```
gcc -O3 -fopenmp mpfr_scan.c /lib/x86_64-linux-gnu/libmpfr.so.6 \
    -lm -o mpfr_scan

for ((a=134217728; a<536870912; a+=4194304)); do
  b=$((a + 4194304))
  OMP_NUM_THREADS=5 ./mpfr_scan "$a" "$b" 192
done
```

The two global extremal records were rerun individually at 1024 bits. A high-precision independent decimal evaluation of the true logarithmic margins is also reported in Section 13; it is not used in place of the directed MPFR inequalities.

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